DEPARTMENT OF THE AIR FORCE
DEAN OF FACULTY
DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
UNITED STATES AIR FORCE ACADEMY

1. COURSE IDENTIFICATION

Course Number:	ECE 383
Course Title:	Embedded Computer Systems II
Semester Hours:	3.0
Term:	Spring 2027
Department:	Department of Electrical and Computer Engineering
Prerequisites:	ECE 382
Co-requisites:	None

2. COURSE DESCRIPTION

A course in the design of digital systems using microprocessors, special-purpose processors, and field-programmable arrays (FPGAs). Students investigate processor architecture tradeoffs, implementation of high-speed digital circuits, configuration of processors on FPGAs, input/output, peripherals, hardware-software interfacing techniques, memory systems, and bus features, including serial communications. Lab.

SECTIONS 3-9 Not for PA Release

3. COURSE DIRECTOR / INSTRUCTOR INFORMATION

3a. Course Director

Name / Rank:	Dr. George York
Office:	Fairchild, 2E46
Phone:	719-333-3190
Email:	george.york@afacademy.af.edu
Office Hours:	Monday-Friday 0800-1600 or by appointment

3b. Instructors

Instructor:	Dr. George York, Fairchild 2E46, george.york@afacademy.af.edu

4. STUDENT LEARNING OUTCOMES

4a. Course Learning Outcomes

The course goals are that cadets shall (1) design, build, and debug hardware/software to drive peripheral devices; (2) design, build, and debug an advanced digital system, taking three different approaches: a fully custom-built hardware solution, a general-purpose software processor, and a general-purpose software processor combined with custom-built hardware; and (3) express the tradeoffs between choosing a microcontroller versus a custom digital system. Upon successful completion of this course, cadets will be able to:

1. Design and implement a complex, structured, synthesizable digital system that meets defined requirements.
2. Use contemporary software tools to debug a digital system design and verify that a digital system meets defined requirements.
3. Apply structured design techniques to design functional hardware interfaces and to provide the basis for efficient, modular software programs.
4. Analyze and describe the timing, clock, and synchronization requirements for a given digital system.
5. Design and implement a digital system using (a) a fully custom-built hardware solution, (b) a general-purpose software processor, and (c) a general-purpose software processor combined with custom-built hardware.
6. Evaluate, analyze, and modify a given digital system to improve area, power, or speed.
7. Express the tradeoffs between choosing a microcontroller versus a custom digital system.

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5. REQUIRED MATERIALS

#	Title / Description	Author(s)	Edition / Notes
1	*RTL Hardware Design Using VHDL: Coding for Efficiency, Portability, and Scalability* (ISBN 0471720925). Free on O'Reilly Books through DoD Libraries (daf.dodmwrlibraries.org).	Pong P. Chu	Course text
2	GitHub account and repository	GitHub	Required; cadet software repository for labs

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6. GRADING

6a. Assignments/Course Assessments

Grading is primarily performed in Gradescope, with official grades listed in Blackboard. The grade distribution for this course is:

Component	Prog (pts)	Final (pts)
Graded Reviews (GRs)	*40*	*25*
Lab 1	*40*	*10*

Lab 2	*0*	*10*
Lab 3	*–*	*10*
Lab 4	*–*	*10*
Quizzes / CPH	*20*	*10*
Final Project	*–*	*25*
Subjective	*–*	*–*
TOTAL	***100***	**100**

6b. Grading Scale

Electrical and Computer Engineering courses are contract graded using the following 100-point scale.

A	A-	B+	B	B-	C+	C	C-	D	F
100%-93%	*92% - 90%*	*89% - 87%*	*86% - 83%*	*82% - 80%*	*79% - 77%*	*76% - 73%*	*72% - 70%*	*69% - 60%*	*59% - 0%*

6c. Assignment Description

Graded Reviews (GRs)	All exams are closed textbook and notes. Both laboratory and classroom work will appear on exams. GRs are individual efforts (see Collaboration and Graded Review Make-ups below).
Labs 1–4	The labs in this course are very hardware intensive. A disciplined approach to design, implementation, and testing is key to success; the "Big Bang" method (wiring and coding something up and debugging it until it works) usually does not work. Incomplete labs may be turned in, but up to 25% can be deducted for not meeting requirements (see the late work policy). Cadets use GitHub to provide their source code for labs. Students must complete all labs, meeting at least ”basic functionality.” Failure to meet ”basic functionality” in all labs will result in a ”controllable incomplete” grade, until all labs are complete.
Quizzes	Cadets who miss a lesson remain responsible for any quiz that lesson. Quizzes may be announced or not announced prior to the lesson.
CPH - Computer Homeworks	Graded primarily in Gradescope. These assignments are a combination of Preflights (what you should have learned preparing for the upcoming lesson) and homework (problems based on the last lesson). Cadets who miss a lesson remain responsible for any assignment due that lesson.
Final Project	Details on the final project are on the class website. Each student must do a different final project, approved by the instructor.
Subjective	Listed in the grade distribution with no points assigned.

7. COURSE POLICIES

Attendance	For CAS notification, email your instructor prior to your absence and include the lesson number, the date, and the reason (a descriptive reason; do not just send a CAS code or SCA number) as soon as possible, preferably before the absence occurs. It is your responsibility to check your SCA to see if instructor permission is required; if it is, you must make the request prior to your absence. If you miss class, you are responsible for all material (e.g., assignments, notes, announcements, handouts) covered in class; check with another cadet in your section to find out what you missed.
Missed Graded Events	When a cadet is absent on the day that an assignment is due, or on the date of a quiz or GR, the cadet is responsible for meeting the following standards: **Scheduled Absence:** If a cadet will miss any graded event due to a scheduled absence such as an SCA, sport team trip, or scheduled LASIK surgery, the cadet is expected to complete all work BEFORE the absence.
	Unscheduled Absence: If a cadet misses a graded event for an unscheduled reason such as AOC-approved bedrest or a family emergency, the cadet must complete all work on the first full class day that they return to duty in order to avoid a late penalty. For example, if a cadet is on AOC bedrest for a GR on M17 and can return to duty on T17 or M18, the cadet is expected to make up the work by M18. **Unique Circumstances:** For circumstances that do not fall under either of these broad categories (e.g., concussion protocol), the cadet is expected to communicate early and often with the instructor. The instructor and Course Director will work with the cadet on a course of action.
Academic Integrity	*See the Cadet Wing Honor Code for details.* All graded work is expected to be your own work. Copying another person's work, with or without documentation, will result in NO academic credit. Furthermore, copying without attribution is dishonorable and will be dealt with as an honor code violation.
Late Work	All work is due as shown on Gradescope. If problems arise with graded assignments, see your instructor in advance. The cutoff for on-time submission is usually 2359 on the due date, but check Gradescope for official due dates and times. late turn-in penalties are: 10% for the first calendar day. 30% for the second calendar day. 60% for the third calendar day. 100% for four or more calendar day (at this point you earn a 0). If notified in advance, an instructor can make individual exceptions to this policy or possibly grant extensions.

EI	Schedule EI with an instructor if you are having difficulty with the course material. You must have read the assignment and attempted the homework before requesting EI. You are responsible for material if you miss class, so get notes from someone in your section. For example, if you miss the lesson where the instructor announces a quiz for the next lesson or assigns homework due next lesson, you are still responsible for the quiz, homework, or any other assignments made. It is in your best interest to check with your classmates after an absence. After you have read the assignment, attempted the homework, and checked with your classmates, you may then schedule EI if you have difficulty with the material, not to make up a class you missed.
Use of AI Tools	You may use ChatGPT and other LLMs to write test cases and help you debug. You may not use them to write source code for you (e.g., VHDL or C/C++).
Documentation Statement	You must document all help received from any sources other than your instructor or instructor-provided course materials (including your textbook). Each documentation statement must be specific enough to explicitly describe what assistance was provided, how it was used to complete the assignment, and who provided the assistance. If no help was received on an assignment, the documentation statement must state "None." If you checked answers with anyone, you must document with whom on which problems. You must document whether you made any changes or not; if you did make changes, you must document the problems you changed and the reasons why. Vague documentation statements will result in a 5% deduction on the assignment. See the sample documentation statements in Section 8. If you use AI tools (see above), you must document this along with including a link to your AI session, which should include the prompts and responses.
Collaboration	Unless specifically directed otherwise (or noted on the assignment), you may collaborate on all assignments with any other cadets currently enrolled in the course. However, all your work and code must be your own. You may ask classmates generic questions like ”how do you write a For Loop”, but not specific code for the assignment. You cannot ask a classmate to debug your code without getting the instructors permission first. GRs are individual efforts; no collaboration is allowed while taking these exams. All electronic devices (phones, smartwatches, computers, tablets, etc.) must be placed out of sight for the duration of the event. If any electronic device is seen during the event, the cadet will receive a zero for that effort. Authorized resources include any material from the course site and online sources regarding VHDL/C/C++ programming syntax only. This does not include any solutions or solution stubs for challenges similar to those asked in any assignments.

Graded Review Make-ups	**Scheduled Absence:** If you know that you will be unable to take the GR during the scheduled GR period, you are required to inform your instructor as soon as possible before the GR to schedule a make-up exam. **Unscheduled Absence:** If you miss the GR for reasons beyond your control (e.g., hospitalization, emergency leave, delayed field trip return), you must contact DFEC (x3190) within two working days to schedule a make-up. Exceptions can only be granted by the Department Head.

8. ADDITIONAL INFORMATION

Course Website and Communication: All communication and course material is provided through a course and section Team.

- **Course website:** usafa-ece.github.io/ece383 has the course schedule (schedule page), course handouts, and course assignments. The website is being updated to the Jupyter Book format used in other DFEC courses; if something is missing or unclear, the previous course website at georgeyork.github.io/ECE383_web may help.

- **Microsoft Teams:** The course Teams site is used for course announcements. Virtual class sessions are held in your section's Teams site when there is no in-person class. Lecture notes will be posted in the Team's channel.

- **Blackboard:** Used only for the "official" grades.

- **Gradescope:** Used to grade most assignments.

- **GitHub:** Your software repository; used to provide your source code for labs.

- **YouTube:** Videos are uploaded to a YouTube channel for your convenience.

Course Philosophy: This course is designed to help in your development as a computer or electrical engineer. As such, economic considerations as well as manufacturability and sustainability are introduced during the first half of the semester and re-emphasized as the semester progresses. Feel free to provide feedback on the lessons and labs at any time; if you have ideas to improve or enhance the course, let your instructor know. The class builds on concepts from the prerequisites, so it is important to seek help as soon as you need it. Procrastination is truly the enemy in a hardware design course. A little foresight and planning and a lot of effort will result in an extremely rewarding experience, serving as the basis for future microprocessor and embedded systems design work.

Sample Documentation: Consider the following examples when writing your own detailed documentation statements:

- *Bad:* Cadet McFly explained how the factorial worked. *Good:* Cadet McFly explained how a recursive function worked conceptually, using diagrams and the assignment materials. He did not look at my code nor did I look at his code during this discussion.

- *Bad:* Cadet McFly helped fix my factorial() method. *Good:* Cadet McFly helped fix my

factorial() method by looking at my code and finding that I had n > 0 instead of n >= 0 on line 85. (A situation such as this may result in less than full credit for the factorial() method, but due to the documentation statement there is no violation of the honor code.)

• *Bad:* Cadet McFly and I worked together on the factorial() method. *Good:* Cadet McFly and I worked together on the factorial() method, each contributing equally to its development. Prior to this help, neither of our factorial() methods was working. My factorial() method is now nearly identical to Cadet McFly's factorial() method. (In a situation such as this, at most half-credit would be earned for the factorial() method, but due to the documentation statement there is no violation of the honor code.)

• *Bad:* Cadet McFly showed me how the factorial() method works. *Good:* Cadet McFly showed me how the factorial() method works by letting me look at his code. Prior to this help, my own factorial() method was not working. My factorial() method is now nearly identical to Cadet McFly's factorial() method. (In a situation such as this, no points would be earned for the factorial() method, but due to the documentation statement there is no violation of the honor code.)

• *Bad:* Cadet McFly showed me how the factorial() method works. *Good:* Cadet McFly showed me how the factorial() method works by looking at my code and talking me through each line as I wrote it. Prior to this help, my own factorial() method was wrong. My factorial() method is now nearly identical to Cadet McFly's factorial() method. (In a situation such as this, no points would be earned for the factorial() method, but due to the documentation statement there is no violation of the honor code.)

9. Course Schedule

Lesson	Topic	Pre-Flight	Graded Event
1	Lec 1: Intro, VHDL, Digital Systems Design	- read 1.1-1.7 - read L1notes - lesson slides on Teams Site	
2	Lec 2: Digital Systems, hierarchical design, testbench	- Homework 1 due BOC - read 2.1, 2.2.1, 2.2.2 - read L2notes - lesson slides on Teams Site	
3	Lec 3: Combinational elements, unsigned, constraints file, synthesis	- Homework 2 due BOC - read 3.5.4, 4.2.3, 4.3.1 - read L3notes - lesson slides on Teams Site	
4	Lec 4: Sequential elements	- Homework 3 due BOC - read 5.1, 5.2, 5.7, 5.8 - read L4notes - lesson slides on Teams Site	
5	Lec 5: Combinations of elements, Lab intro	- Homework 4 due BOC - read 7.2 - read L5notes	
6	Lab1 - VGA Synchronization	- Homework 5 due BOC - Gate Check 1 due taps L6	
7	Lab1 - VGA Synchronization	- Gate Check 2 due taps L7	
8	Lab1 - VGA Synchronization	- Lab1 Functionality due taps L8	
9	Finite State Machines	- Lab1 README due taps L9 - read 10.2.1, 10.3.2, 10.4, 10.6.1 - read L9notes - lesson slides on Teams Site	
10	Datapath and Control	- Homework 6 due BOC - read 11.1, 11.2, 14.4.2 - read L10notes - lesson slides on Teams Site	
11	Datapath and Control	- Homework 7 due BOC - read 11.5 - read L11notes - lesson slides on Teams Site	

12	Datapath and Control, Lab intro	- Homework 8 due BOC - lesson slides on Teams Site	
13	Lab 2: Data acquisition, storage, and display	- Homework 8b2 due BOC	
14	Lab 2: Data acquisition, storage, and display	- Gate Check 1 due BOC	
15	Lab 2: Data acquisition, storage, and display	- Gate Check 2 due BOC	
16	Lab 2: Data acquisition, storage, and display	- Gate Check 3 due BOC	
17	Lab 2: Data acquisition, storage, and display	- functionality due taps L17	
18	GR1	GR1	GR1
19	Soft CPU - do ICE 1	- Lab2 README due BOC - read L19notes - lesson slides on Teams Site	
20	Soft CPU - do ICE 2	- Homework 9 - ICE 1 due BOC - lesson slides on Teams Site	
21	Soft CPU - do ICE 3	- ICE 2 due BOC - read L21notes - Final Project ideas, youtube? - Final Project ideas - lesson slides on Teams Site	
22	Lab 3: Oscilloscope Control	- ICE 3 due BOC	
23	Lab 3: Oscilloscope Control	- Gate Check 1 due BOC	
24	Lab 3: Oscilloscope Control	- Gate Check 2 due BOC	
25	Lab 3: Oscilloscope Control	- Gate Check 3 due taps	
26	Lab 3: Oscilloscope Control	- functionality due taps L26	
27	Direct Digital Synthesis	- Lab3 README due BOC - Final Project Proposal due taps - read L27notes - lesson slides on Teams Site	

28	Direct Digital Synthesis	- Homework 12 due BOC - read L28notes - lesson slides on Teams Site	
29	Lab 4: Function Generator	- Homework 13 due BOC - Gate Check 1 due taps L29	
30	Lab 4: Function Generator	- Gate Check 2 due taps L30	
31	Lab 4: Function Generator	- functionality due taps L31	
32	GR2	GR2	GR2
33	Lab 5: Final Project	- Lab4 README due BOC	
34	Lab 5: Final Project	- Final Project Plan (and revised proposal) due BOC	
35	Lab 5: Final Project		
36	Lab 5: Final Project		
37	Lab 5: Final Project		
38	Lab 5: Final Project		
39	Lab 5: Final Project		
40	Final Project Presentation/Demo Day	- Functionality due BOC - Presentation slides due BOC	Presentation/Demo
41	Final Report	- Final Report due taps L41	

Note: *Lesson dates will follow the Spring 2027 academic calendar.*